

Directions: It's been a while since covered a variety of topics from this course. This review worksheet covers *some* (not necessarily *all*) of the topics that we have learned. **DO NOT** take this review to be a copy of the final exam. Be sure to look over the Final Exam Study Guide and take heed of the other recommendations for preparing for the file. Remember that many of these problems are still *Challenge Problems* and therefore it is not expected that you might know how to answer each of these questions immediately.

1. Consider the region bounded by the curves $y = x^2 + 1$, $y = 1$ and $x = 3$.
 - a) Set up, but do not evaluate, an integral(s) that represents the volume of this solid created by rotating this region about the x -axis by integrating with respect to x .

When working with respect x here we will be using washers.

$$V = \int_0^3 \left[\pi(x^2 + 1 - 0)^2 - \pi(1 - 0)^2 \right] dx$$

- b) Set up, but do not evaluate, an integral(s) that represents the volume of this solid created by rotating this region about the x -axis by integrating with respect to y .

When working with respect y here we will be using shells.

$$V = \int_1^{10} 2\pi y (3 - \sqrt{y-1}) dy$$

- c) Consider the solid formed with this region as its base and cross sections parallel to the x -axis are regular hexagons. Set up, but do not evaluate, an integral(s) that represents the volume of this solid.

The area formula for a regular hexagon is $A = \frac{3\sqrt{3}}{2} a^2$, where a represents the length of a side of the hexagon. (Note that it is not expected that you would know this formula and would provided a formula like this in an exam.)

The length of the side a is given by $a = 3 - \sqrt{y-1}$. Therefore, the total volume of the resulting solid is

$$V = \int_1^{10} \frac{3\sqrt{3}}{2} (3 - \sqrt{y-1})^2 dy$$

2. A bullet is formed by revolving the area bounded by the curve $y = \ln(x)$ from $x = 1$ to $x = e$ about the x axis. It is then shot straight into a very thick wall and does not pierce through the other side. As such, it makes a cylindrical hole until it stops moving. Then the bullet is carefully extracted with affecting the hole at all, leaving an empty hole with a pointy end where the bullet once was. The length of the entire hole is $e+1$. Determine the volume of the hole created by the bullet.

First note that the region that must be revolved to create the bullet has the following picture.

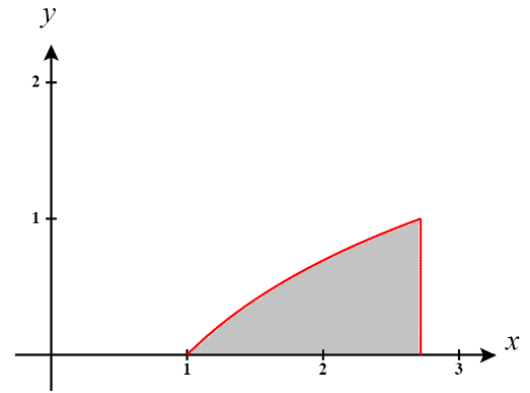
This implies that the volume of the bullet is given by

$$V = \int_1^e \pi (\ln(x))^2 dx = \pi \int_1^e (\ln(x))^2 dx$$

Let $u = (\ln(x))^2$ & $dv = dx$, then $du = 2\ln(x) \frac{1}{x} dx$ & $v = x$.

Then,

$$\begin{aligned} \pi \int_1^e (\ln(x))^2 dx &= \pi \left[x \ln(x) \Big|_1^e - \int_1^e x 2 \frac{1}{x} \ln(x) dx \right] \\ &= \pi \left[(e \ln(e)) - (1 \ln(1)) - \int_1^e 2 \ln(x) dx \right] \\ &= \pi \left[e - 0 - 2 \left[x \ln(x) - x \right] \Big|_1^e \right] \\ &= \pi \left[e - 2 \left[(e \ln(e) - e) - (1 \ln(1) - 1) \right] \right] \\ &= \pi \left[e - 2 \left[(e - e) - 0 + 1 \right] \right] \\ &= \pi [e - 2] \end{aligned}$$



Note that this bullet has a length of $e - 1$ and if the total length of the entire hole was $e + 1$ then this implies that the depth of the empty cylindrical space must be 2. We can also see that radius of the hole created by the bullet would be the same as the widest part of the bullet which is 1.

Therefore, the volume of the hole is given by

$V = \text{volume of space where bullet was} + \text{volume of cylindrical area}$

$$\begin{aligned} &= \pi [e - 2] + \pi (1)^2 (2) \\ &= \pi [e - 2 + 2] \\ &= \pi e \end{aligned}$$

3. Compute each of the following integrals.

a) $\int e^{\sqrt{x}} dx$ (Hint: Start by substituting $x = y^2$)

If we substitute $x = y^2 \rightarrow \sqrt{x} = y$ then we have that $2ydy = dx$.

Therefore, $\int e^{\sqrt{x}} dx = \int e^y 2y dy$.

Now let $u = 2y$ & $dv = e^y dy$. This implies that $du = 2dy$ & $v = e^y$. By IBP we have that

$$\int e^y 2y dy = 2ye^y - \int 2e^y dy = 2ye^y - 2e^y + C = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

b) $\int \frac{dx}{(5+2x+x^2)^{3/2}}$

We proceed by completing the square here.

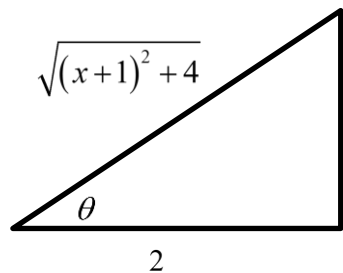
$$\int \frac{dx}{(5+2x+x^2)^{3/2}} = \int \frac{dx}{((x^2+2x+1)+5-1)^{3/2}} = \int \frac{dx}{((x+1)^2+4)^{3/2}}.$$

Now we attempt a trig substitution.

Let $x+1 = 2 \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, then $dx = 2 \sec^2(\theta) d\theta$.

$$\begin{aligned} \int \frac{dx}{((x+1)^2+4)^{3/2}} &= \int \frac{2 \sec^2(\theta)}{(4 \tan^2(\theta)+4)^{3/2}} d\theta = \int \frac{2 \sec^2(\theta)}{8 (\sec^2(\theta))^{3/2}} d\theta = \int \frac{\sec^2(\theta)}{4 |\sec(\theta)|^3} d\theta \\ &= \int \frac{\sec^2(\theta)}{4 \sec^3(\theta)} d\theta = \int \frac{1}{4} \cos(\theta) d\theta = \frac{1}{4} \sin(\theta) + C \end{aligned}$$

Since $x+1 = 2 \tan(\theta)$ we know that $\tan(\theta) = \frac{x+1}{2}$ and can set up the following reference triangle.



Therefore, $\frac{1}{4} \sin(\theta) + C = \frac{1}{4} \cdot \frac{x+1}{\sqrt{(x+1)^2 + 4}} + C = \frac{x+1}{4\sqrt{x^2 + 2x + 5}} + C$

c) $\int \frac{\sin^3(\theta)}{\cos^2(\theta)} d\theta$

$$\begin{aligned} \int \frac{\sin^3(\theta)}{\cos^2(\theta)} d\theta &= \int \sin(\theta) \frac{\sin^2(\theta)}{\cos^2(\theta)} d\theta = \int \sin(\theta) \tan^2(\theta) d\theta = \int \sin(\theta) [\sec^2(\theta) - 1] d\theta \\ &= \int [\sin(\theta) \sec^2(\theta) - \sin(\theta)] d\theta = \int [\tan(\theta) \sec(\theta) - \sin(\theta)] d\theta \\ &= \sec(\theta) + \cos(\theta) + C \end{aligned}$$

d) $\int \frac{x^3 + x - 1}{x^2 - 1} dx$

Since we have an improper rational expression we have to start with a polynomial long division. We calculate that

$$\begin{array}{r} x \\ (x^2 - 1) \overline{) x^3 + 0x^2 + x - 1} \\ \underline{-(x^3 + 0x^2 - x)} \\ 2x - 1 \end{array} \quad \text{This implies that } \frac{x^3 + x - 1}{x^2 - 1} = x + \frac{2x - 1}{x^2 - 1}.$$

Therefore $\int \frac{x^3 + x - 1}{x^2 - 1} dx = \int \left(x + \frac{2x - 1}{x^2 - 1} \right) dx = \int \left(x + \frac{A}{x + 1} + \frac{B}{x - 1} \right) dx$

We need $2x - 1 = A(x - 1) + B(x + 1)$.

Note that if $x = 1$ then $1 = 2B \rightarrow B = \frac{1}{2}$

Note that if $x = -1$ then $-3 = -2A \rightarrow A = \frac{3}{2}$

So, we have that $\int \left(x + \frac{A}{x+1} + \frac{B}{x-1} \right) dx = \int \left(x + \frac{3/2}{x+1} + \frac{1/2}{x-1} \right) dx = \frac{1}{2}x^2 + \frac{3}{2}\ln|x+1| + \frac{1}{2}\ln|x-1| + C$

4. The volume of the solid formed by revolving the curve $y = \frac{1}{\sqrt{x}} - \frac{1}{\sqrt{1+x}}$ bounded between $x = \frac{1}{8}$ and $x = \frac{9}{16}$ around the x -axis can be expressed in the form $A\pi \ln\left(\frac{B}{C}\right)$. If A and B contain no factors that are perfect squares, then find the value of $A \times B \times C$.

$$\begin{aligned} V &= \int_{1/8}^{9/16} \pi \left[\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{1+x}} \right]^2 dx = \int_{1/8}^{9/16} \pi \left[\frac{1}{x} - \frac{2}{\sqrt{x}\sqrt{1+x}} + \frac{1}{1+x} \right] dx \\ &= \int_{1/8}^{9/16} \pi \left[\frac{1}{x} - \frac{2}{\sqrt{x+x^2}} + \frac{1}{1+x} \right] dx = \int_{1/8}^{9/16} \pi \left[\frac{1}{x} - \frac{2}{\sqrt{\left(x^2+x+\frac{1}{4}\right) - \frac{1}{4}}} + \frac{1}{1+x} \right] dx \\ &= \int_{1/8}^{9/16} \pi \left[\frac{1}{x} - \frac{2}{\sqrt{\left(x+\frac{1}{2}\right)^2 - \frac{1}{4}}} + \frac{1}{1+x} \right] dx = \int_{1/8}^{9/16} \pi \left[\frac{1}{x} + \frac{1}{1+x} \right] dx - \int_{1/8}^{9/16} \frac{2\pi}{\sqrt{\left(x+\frac{1}{2}\right)^2 - \frac{1}{4}}} dx \end{aligned}$$

On the second integral we let $x + \frac{1}{2} = \frac{1}{2}\sec(\theta)$ and then $dx = \frac{1}{2}\sec(\theta)\tan(\theta)$ where $0 \leq \theta < \frac{\pi}{2}$ or

$\pi \leq \theta < \frac{3\pi}{2}$. Our bounds become $\frac{1}{8} + \frac{1}{2} = \frac{1}{2}\sec(\theta) \rightarrow \sec(\theta) = \frac{5}{4}$ and

$\frac{9}{16} + \frac{1}{2} = \frac{1}{2}\sec(\theta) \rightarrow \sec(\theta) = \frac{17}{8}$. Clearly since $\sec(\theta) > 0$ on this interval we can further restrict our

angle restriction to be $0 \leq \theta < \frac{\pi}{2}$ and therefore we can assert that we have $\theta = \sec^{-1}\left(\frac{5}{4}\right)$ and

$\theta = \sec^{-1}\left(\frac{17}{8}\right)$ as bounds for this integral. Then we have that

$$\begin{aligned}
& \int_{1/8}^{9/16} \pi \left[\frac{1}{x} + \frac{1}{1+x} \right] dx + \int_{1/8}^{9/16} \frac{2\pi}{\sqrt{\left(x + \frac{1}{2}\right)^2 - \frac{1}{4}}} dx \\
&= \pi \left[\ln|x| + \ln|1+x| \right]_{1/8}^{9/16} - \int_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} \frac{2\pi}{\sqrt{\left(\frac{1}{2}\sec(\theta)\right)^2 - \frac{1}{4}}} \frac{1}{2} \sec(\theta) \tan(\theta) d\theta \\
&= \pi \left[\ln\left(\frac{9}{16}\right) + \ln\left(1 + \frac{9}{16}\right) \right] - \pi \left[\ln\left(\frac{1}{8}\right) + \ln\left(1 + \frac{1}{8}\right) \right] - \int_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} \frac{\pi \sec(\theta) \tan(\theta)}{\frac{1}{2} \sqrt{\sec^2(\theta) - 1}} d\theta \\
&= \pi \left[\ln\left(\frac{9}{16}\right) + \ln\left(\frac{25}{16}\right) \right] - \pi \left[\ln\left(\frac{1}{8}\right) + \ln\left(\frac{9}{8}\right) \right] - \int_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} \frac{2\pi \sec(\theta) \tan(\theta)}{|\tan(\theta)|} d\theta \\
&= \pi \ln\left(\frac{225}{256}\right) - \pi \ln\left(\frac{9}{64}\right) - \int_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} \frac{2\pi \sec(\theta) \tan(\theta)}{\tan(\theta)} d\theta \\
&= \pi \ln\left(\frac{225/256}{9/64}\right) - \int_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} 2\pi \sec(\theta) d\theta \\
&= \pi \ln\left(\frac{25}{4}\right) - 2\pi \int_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} \sec(\theta) d\theta \\
&= \pi \ln\left(\frac{25}{4}\right) - 2\pi \ln|\sec(\theta) + \tan(\theta)|_{\sec^{-1}\left(\frac{5}{4}\right)}^{\sec^{-1}\left(\frac{17}{8}\right)} \\
&= \pi \ln\left(\frac{25}{4}\right) - 2\pi \ln\left|\frac{17}{8} + \frac{15}{8}\right| + 2\pi \ln\left|\frac{5}{4} + \frac{3}{4}\right| \\
&= \pi \ln\left(\frac{25}{4}\right) - 2\pi \ln\left(\frac{32}{8}\right) + 2\pi \ln\left(\frac{8}{4}\right) = \pi \ln\left(\frac{25}{4}\right) - 2\pi \ln(4) + 2\pi \ln(2) \\
&= \pi \ln\left(\frac{25}{4}\right) + 2\pi \ln\left(\frac{1}{2}\right) = \pi \ln\left(\frac{25}{4}\right) + \pi \ln\left(\frac{1}{4}\right) \\
&= \pi \ln\left(\frac{25}{16}\right) = 2\pi \ln\left(\frac{5}{4}\right)
\end{aligned}$$

So, our answer is in the form $A\pi \ln\left(\frac{B}{C}\right)$ where A and B are numbers with no square factors where $A=2$, $B=5$, & $C=4$. We see that $A \times B \times C = (2)(5)(4) = 40$.

5. Evaluate the integral $\int_{-1}^1 \frac{1}{x} dx$.

We see that $f(x) = \frac{1}{x}$ has a discontinuity at $x=0$ and therefore this is an improper integral. We rewrite

this integral as
$$\int_{-1}^1 \frac{1}{x} dx = \int_{-1}^0 \frac{1}{x} dx + \int_0^1 \frac{1}{x} dx = \lim_{t \rightarrow 0^-} \left[\int_{-1}^t \frac{1}{x} dx \right] + \lim_{s \rightarrow 0^+} \left[\int_s^1 \frac{1}{x} dx \right].$$

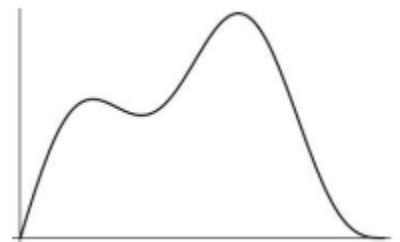
We see that $\lim_{t \rightarrow 0^-} \left[\int_{-1}^t \frac{1}{x} dx \right] = \lim_{t \rightarrow 0^-} \left[\ln|x| \right]_{-1}^t = \lim_{t \rightarrow 0^-} [\ln|t| - \ln(1)] = -\infty$. Since this limit doesn't exist the entire integral is divergent.

6. Consider the curve $y = 2\sin(x) + \sin(2x)$, $0 \leq x \leq \pi$ shown at the right.

- a) Set up, but do not evaluate, an integral which represents the length of the curve.

Note that $y' = 2\cos(x) + 2\cos(2x)$ is continuous on $[0, \pi]$ and therefore

$$L = \int_0^{\pi} \sqrt{1 + [2\cos(x) + 2\cos(2x)]^2} dx$$



- b) Set up, but do not evaluate, an integral which represents surface area obtained by rotating the curve about the x -axis.

Note that $y' = 2\cos(x) + 2\cos(2x)$ is continuous on $[0, \pi]$ and y is nonnegative on $[0, \pi]$. Therefore,

$$S = \int_0^{\pi} 2\pi \left[2\sin(x) + \sin(2x) \right] \sqrt{1 + \left[2\cos(x) + 2\cos(2x) \right]^2} dx$$

- c) Set up, but do not evaluate, an integral which represents surface area obtained by rotating the curve about the y -axis.

This integral is typically set up with respect to y but this will not be possible here since we cannot solve the given equation for x . Therefore, we will still set this integral up with respect to x .

$$S = \int_0^{\pi} 2\pi x \sqrt{1 + \left[2\cos(x) + 2\cos(2x) \right]^2} dx$$

7. Recall that the Monotone Sequence Theorem test states that if a sequence a_n is (i) *increasing* and (ii) *bounded* then it must converge. Consider the sequence

$$a_1 = \sqrt{2}$$

$$a_2 = \sqrt{2 + \sqrt{2}}$$

$$a_3 = \sqrt{2 + \sqrt{2 + \sqrt{2}}}$$

...

$$a_{n+1} = \sqrt{2 + a_n}$$

- a) Show that if $a_n < 2$ then $a_{n+1} < 2$. Since $a_1 = \sqrt{2} < 2$ what does this implies about all values of a_n ?

If $a_n < 2$, then $a_{n+1} = \sqrt{2 + a_n} < \sqrt{2 + 2} = 2 \rightarrow a_{n+1} < 2$.

So, since $a_1 < 2$ and whenever $a_n < 2$ it is the case that $a_{n+1} < 2$ then it must be the case that $a_n < 2$ for all values of n .

This means that a_n is bounded above by 2.

- b) Show that a_n increasing.

$\frac{a_{n+1}}{a_n} = \frac{\sqrt{2 + a_n}}{a_n} = \frac{\sqrt{2 + a_n}}{\sqrt{a_n^2}} = \sqrt{\frac{2 + a_n}{a_n^2}} > \sqrt{\frac{a_n + a_n}{a_n^2}} = \sqrt{\frac{2}{a_n}} > \sqrt{\frac{2}{2}} = 1$. Therefore $\frac{a_{n+1}}{a_n} > 1 \rightarrow a_{n+1} > a_n$ for all values of n and thus a_n is increasing.

- c) Conclude that a_n must converge. Can you see what it must converge to?

Since a_n is bounded above and increasing, by the MST, a_n must converge.

Suppose that $\lim_{n \rightarrow \infty} a_n = L$ then it must also be the case that $\lim_{n \rightarrow \infty} a_{n+1} = L$.

This implies that $L = \lim_{n \rightarrow \infty} \sqrt{2 + a_n} = \sqrt{2 + \lim_{n \rightarrow \infty} a_n} = \sqrt{2 + L}$

Which implies that

$$\begin{aligned} L = \sqrt{2 + L} &\rightarrow L^2 - L - 2 = 0 \\ &\rightarrow (L - 2)(L + 1) = 0 \\ &\rightarrow L = 2 \text{ or } L = -1 \end{aligned}$$

Clearly $L = -1$ is an extraneous solution and so it must be the case that $L = 2$.

8. Determine the limit of the sequence $a_n = (-1)^n \arctan(n)$.

We see that for even numbered terms of the sequence we have $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \arctan(n) = \frac{\pi}{2}$.

For odd numbered terms of the sequence we have that $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} -\arctan(n) = -\frac{\pi}{2}$

Since the even and odd numbered terms don't head towards the same value the limit of the given sequence *DNE*.

9. Determine if the following series converges or diverges $\sum_{n=2}^{\infty} (-1)^n n e^{-n}$

The given series is an alternating series with $a_n = n e^{-n} = \frac{n}{e^n}$. Notice that if $f(x) = \frac{x}{e^x}$ then

$f'(x) = \frac{e^x - x e^x}{(e^x)^2} = \frac{1 - x}{e^x}$ and for $x \geq 2$ it is the case that $f'(x) \leq 0$ which means that $f(x)$ is decreasing

on $[2, \infty)$. This implies that a_n is decreasing. We also see that $\lim_{n \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x}{e^x} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$ which implies, by the AST, that the given series converges.

10. Determine if the each of the following series is absolutely convergent, conditionally convergent, or divergent.

a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

Note that the given series contains only non-zero terms. We calculate that

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{(n+1)^2}{2^{n+1}} \right|}{\left| \frac{n^2}{2^n} \right|} = \lim_{n \rightarrow \infty} \frac{(n+1)^2 2^n}{2^{n+1} n^2} = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^2} = \frac{1}{2} < 1.$$

Therefore, by the Ratio Test, the given series converges absolutely.

b) $\sum_{n=2}^{\infty} (-1)^n \frac{n^2}{n^3 - 1}$

Note that $\sum_{n=2}^{\infty} \left| (-1)^n \frac{n^2}{n^3 - 1} \right| = \sum_{n=2}^{\infty} \frac{n^2}{n^3 - 1}$. We compare this series with $\sum_{n=2}^{\infty} \frac{1}{n}$ which is the Harmonic series with a single term missing and therefore still diverges. Note that both series contain only positive terms

and $\lim_{n \rightarrow \infty} \frac{\left(\frac{n^2}{n^3 - 1} \right)}{\left(\frac{1}{n} \right)} = \lim_{n \rightarrow \infty} \frac{n^3}{n^3 - 1} = 1$. Therefore, by the LCT both series must have the same behavior.

Since $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges, then $\sum_{n=2}^{\infty} \frac{n^2}{n^3 - 1}$ must also diverge. Therefore, the given series is absolutely divergent.

The given series is an alternating series with $a_n = \frac{n^2}{n^3 - 1}$. Notice that if $f(x) = \frac{x^2}{x^3 - 1}$ then

$$f'(x) = \frac{(x^3 - 1)2x - x^2(3x^2)}{(x^3 - 1)^2} = \frac{-4x^4 - 2x}{(x^3 - 1)^2} \text{ and for } x \geq 2 \text{ it is the case that } f'(x) \leq 0 \text{ which means that}$$

$f(x)$ is decreasing on $[2, \infty)$. This implies that a_n is decreasing. We also see that

$\lim_{n \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2}{x^3 - 1} \cdot \frac{1/x^3}{1/x^3} = \lim_{x \rightarrow \infty} \frac{1/x}{1 - 1/x^3} = \frac{0}{1 - 0} = 0$ which implies, by the AST, that the given series converges. Therefore, the given series conditionally converges.

c) $\pi^2 + \pi + 1 + \frac{1}{\pi} + \dots$

We can express this series in sigma notation as $\sum_{n=1}^{\infty} \pi^2 \left(\frac{1}{\pi} \right)^{n-1}$. This is a geometric series with $|r| = \frac{1}{\pi} < 1$ and therefore converges. Note that $\sum_{n=1}^{\infty} \left| \pi^2 \left(\frac{1}{\pi} \right)^{n-1} \right| = \sum_{n=1}^{\infty} \pi^2 \left(\frac{1}{\pi} \right)^{n-1}$ and therefore the series converges absolutely.

d) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{\ln(n+1)}$

Let $f(x) = \frac{\sqrt{x}}{\ln(x+1)}$. Note that $\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\ln(x+1)} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{x+1}{2\sqrt{x}} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1}{\left(\frac{1}{\sqrt{x}} \right)} = \lim_{x \rightarrow \infty} \sqrt{x} = \infty$. This

implies that $\lim_{n \rightarrow \infty} a_n = \infty$ and therefore, by the Test for Divergence, the given series must diverge.

11. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms and that $\sum b_n$ converges.

a) Show that if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ then $\sum_{n=1}^{\infty} a_n$ also converges. (*Hint: Start by showing that the inequality $a_n \leq b_n$ is eventually true*). (Note that this question is exploring an extension of the Limit Comparison Test.)

Since $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$, there exists N such that $\left| \frac{a_n}{b_n} - 0 \right| < 1$ for some $n \geq N$. Therefore $a_n < b_n$ for some $n \geq N$.

By the Comparison Test, $\sum a_n$ must converge since $\sum b_n$ converges.

b) Use part (a) to show that $\sum_{n=1}^{\infty} \frac{[\ln(n)]^2}{n^3}$ converges.

Let $b_n = \frac{1}{n^2}$.

Then, $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{[\ln(n)]^2}{n^3} \cdot \frac{n^2}{1} = \lim_{n \rightarrow \infty} \frac{[\ln(n)]^2}{n}$. Let $f(x) = \frac{[\ln(x)]^2}{x}$. We see that

$$\lim_{x \rightarrow \infty} f(x) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{2 \ln(x) \cdot \frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{2 \ln(x)}{x} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{2}{x} = 0. \text{ Therefore } \lim_{n \rightarrow \infty} \frac{[\ln(n)]^2}{n} = 0$$

So we have that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$.

We also see that $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ which is a convergent p-series (with $p = 2 > 1$).

So, by part (a), $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{[\ln(n)]^2}{n^3}$ must converge.

- c) Give an example of a pair of series $\sum a_n$ and $\sum b_n$ with positive terms where $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ diverges, but $\sum a_n$ converges.

Consider the series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ which is a convergent p-series (with $p = 2 > 1$) and

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ which is the Harmonic series and therefore divergent.

Note that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \frac{n}{1} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

12. Find the Taylor series (centered at the given point) for each of the following functions by differentiating the given functions and finding the pattern.

- a) $f(x) = \sin(x)$ centered at $x_0 = \frac{\pi}{2}$

We want to find: $\sum_{n=0}^{\infty} \frac{f^{(n)}\left(\frac{\pi}{2}\right)}{n!} \left(x - \frac{\pi}{2}\right)^n$ which means we need to determine $f^{(n)}\left(\frac{\pi}{2}\right)$.

We note that:

$$f(x) = \sin(x) \quad \rightarrow \quad f\left(\frac{\pi}{2}\right) = 1$$

$$f'(x) = \cos(x) \quad \rightarrow \quad f'\left(\frac{\pi}{2}\right) = 0$$

$$f''(x) = -\sin(x) \quad \rightarrow \quad f''\left(\frac{\pi}{2}\right) = -1$$

$$f'''(x) = -\cos(x) \quad \rightarrow \quad f'''\left(\frac{\pi}{2}\right) = 0$$

...

So we see that for even values of n , $f^{(n)}\left(\frac{\pi}{2}\right) = \pm 1$, and for odd values of n , $f^{(n)}\left(\frac{\pi}{2}\right) = 0$.

Therefore, the Taylor series is: $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(x - \frac{\pi}{2}\right)^{2n}$

b) $f(x) = e^{3x}$ centered at $x_0 = 2$

We want to find: $\sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!} (x-2)^n$ which means we need to determine $f^{(n)}(2)$.

We note that:

$$f(x) = e^{3x} \rightarrow f(2) = e^6$$

$$f'(x) = 3e^{3x} \rightarrow f'(2) = 3e^6$$

$$f''(x) = 9e^{3x} \rightarrow f''(2) = 9e^6$$

$$f'''(x) = 27e^{3x} \rightarrow f'''(2) = 27e^6$$

...

So we see, $f^{(n)}(2) = 3^n e^6$.

Therefore, the Taylor series is: $\sum_{n=0}^{\infty} \frac{3^n e^6}{n!} (x-2)^n$

13. Find the Radius and Interval of Convergence of the following power series:

a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

By the Ratio Test we see that $\lim_{n \rightarrow \infty} \frac{\left| \frac{x^{n+1}}{n+1} \right|}{\left| \frac{x^n}{n} \right|} = \lim_{n \rightarrow \infty} \frac{n|x|^{n+1}}{n+1|x|^n} = \lim_{n \rightarrow \infty} \frac{n}{n+1} \cdot |x| = |x| \lim_{n \rightarrow \infty} \frac{n}{n+1} = |x|$

Therefore, the series will converge if $|x| < 1 \rightarrow -1 < x < 1$.

We now must check for convergence at the endpoints of the interval.

If $x = -1$ then our series becomes $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ which, by the AST, is a convergent series.

If $x = 1$ then our series becomes $\sum_{n=1}^{\infty} \frac{(1)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$ which is the Harmonic series and diverges.

Therefore, the Interval of Convergence is $[-1, 1)$.

Therefore, the Radius of Convergence is $R = 1$.

b) $\sum_{n=0}^{\infty} \sqrt{n} x^n$

By the Ratio Test we see that

$$\lim_{n \rightarrow \infty} \frac{|\sqrt{n+1} x^{n+1}|}{|\sqrt{n} x^n|} = \lim_{n \rightarrow \infty} \frac{\sqrt{n+1} |x|^{n+1}}{\sqrt{n} |x|^n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n+1}}{\sqrt{n}} \cdot |x| = |x| \lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{n}} = |x| \cdot 1 = |x|$$

Therefore, the series will converge if $|x| < 1 \rightarrow -1 < x < 1$.

We now must check for convergence at the endpoints of the interval.

If $x = -1$ then our series becomes $\sum_{n=1}^{\infty} \sqrt{n} (-1)^n$. We see that when n is even the terms of the series look like $a_n = \sqrt{n}$ and $\lim_{n \rightarrow \infty} \sqrt{n} = \infty$. If n is odd the terms look like $a_n = -\sqrt{n}$ and $\lim_{n \rightarrow \infty} -\sqrt{n} = -\infty$. Therefore, $\lim_{n \rightarrow \infty} a_n$ DNE and so by the Test for Divergence this series diverges.

If $x = 1$ then our series becomes $\sum_{n=1}^{\infty} \sqrt{n} (1)^n = \sum_{n=1}^{\infty} \sqrt{n}$ which diverges by the Test for Divergence since $\lim_{n \rightarrow \infty} \sqrt{n} = \infty$.

Therefore, the Interval of Convergence is $(-1, 1)$.

Therefore, the Radius of Convergence is $R = 1$.

c) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

By the Ratio Test we see that

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{x^{n+1}}{(n+1)!} \right|}{\left| \frac{x^n}{n!} \right|} = \lim_{n \rightarrow \infty} \frac{n! |x|^{n+1}}{(n+1)! |x|^n} = \lim_{n \rightarrow \infty} \frac{1}{(n+1)} \cdot |x| = |x| \cdot \lim_{n \rightarrow \infty} \frac{1}{n+1} = |x| \cdot 0 = 0 < 1$$

Therefore, the series will converge for all values of x .

Therefore, the Interval of Convergence is $(-\infty, \infty)$.

Therefore, the Radius of Convergence is $R = \infty$.

d) $\sum_{n=0}^{\infty} \frac{(-1)^n (x-3)^n}{2n+1}$

By the Ratio Test we see that

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{(-1)^{n+1} (x-3)^{n+1}}{2(n+1)+1} \right|}{\left| \frac{(-1)^n (x-3)^n}{2n+1} \right|} = \lim_{n \rightarrow \infty} \frac{(2n+1)|x-3|^{n+1}}{(2n+3)|x-3|^n} = \lim_{n \rightarrow \infty} \frac{2n+1}{2n+3} \cdot |x-3| = |x-3| \lim_{n \rightarrow \infty} \frac{2n+1}{2n+3} = |x-3|$$

Therefore, the series will converge if $|x-3| < 1 \rightarrow -1 < x-3 < 1 \rightarrow 2 < x < 4$.

We now must check for convergence at the endpoints of the interval.

If $x=2$ then our series becomes

$$\sum_{n=0}^{\infty} \frac{(-1)^n (2-3)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n (-1)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1 \cdot -1)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{(1)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{1}{2n+1}$$

Let $f(x) = \frac{1}{2x+1}$ and note that $f(x)$ is continuous, positive, and decreasing continuous on $[0, \infty)$. We see that

$$\int_0^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \left[\int_0^t \frac{1}{2x+1} dx \right] = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln |2x+1| \Big|_0^t \right] = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln |2t+1| - \frac{1}{2} \ln(1) \right] = \infty \text{ and so this}$$

integral diverges and therefore, by the Integral Test, the series diverges.

If $x=4$ then our series becomes $\sum_{n=0}^{\infty} \frac{(-1)^n (4-3)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n (1)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$. This series clearly converges by the AST.

Therefore, the Interval of Convergence is $(2, 4]$.

Therefore, the Radius of Convergence is $R=1$.

14. Give the Taylor polynomial of degree 2 centered at 4 for $f(x) = \frac{1}{\sqrt{x}}$. Use Taylor's Theorem to estimate the accuracy this approximation for x between 3.8 and 4.2.

Note that $f'(x) = -\frac{1}{2}x^{-2/3}$ and $f''(x) = \frac{3}{4}x^{-5/2}$.

By Taylor's Theorem we know that $|R_2(x)| \leq \frac{M}{(2+1)!} |x-4|^{2+1}$.

We need $|f'''(x)| \leq M$ for all x values within $[3.8, 4.2]$. Note that $f'''(x) = -\frac{15}{8}x^{-7/2}$ and on the interval $[3.8, 4.2]$ it must be the case that $-\frac{15}{8x^{7/2}} \leq \frac{15}{8(3.8)^{7/2}}$. Therefore, we let $M = \frac{15}{8(3.8)^{7/2}}$.

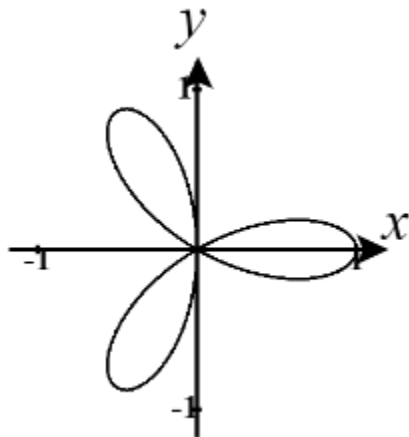
Also, since $3.8 \leq x \leq 4.2 \rightarrow -0.2 \leq x-4 \leq 0.2 \rightarrow |x-4| \leq 0.2$.

Therefore, for x -values in $[3.8, 4.2]$ we have that

$$|R_2(x)| \leq \frac{\left(\frac{15}{8(3.8)^{7/2}}\right)}{(3)!} |x-4|^3 \leq \frac{5}{16(3.8)^{7/2}} (0.2)^3 \approx 0.0000234$$

15. Consider the curve $r = \cos(3\theta)$

a) Make a sketch of this curve and find the total area enclosed by the curve.



b) Set up an integral for the total length of the curve.

We first note that this curve is traced out for $0 \leq \theta \leq \pi$. Therefore, the length of the curve is given by

$$L = \int_0^{\pi} \sqrt{\cos^2(3\theta) + (-3\sin(3\theta))^2} d\theta = \int_0^{\pi} \sqrt{\cos^2(3\theta) + 9\sin^2(3\theta)} d\theta = \int_0^{\pi} \sqrt{1 + 8\sin^2(3\theta)} d\theta$$

c) Determine the total area enclosed by this polar curve.

$$A = \int_0^{\pi} \frac{1}{2} \cos^2(3\theta) d\theta = \int_0^{\pi} \frac{1}{4} [1 + \cos(6\theta)] d\theta = \frac{1}{4} \left[\theta + \frac{1}{6} \sin(6\theta) \right]_0^{\pi} = \frac{1}{4} [\pi + 0] - \frac{1}{4} [0 + 0] = \frac{\pi}{4}$$

16. The coordinates of a dot P moving in the xy -plane at time t are given by:

$$x = 5e^{-t} \cos(t) \text{ \& } y = 5e^{-t} \sin(t)$$

Let L_a be the distance traveled by P for $0 \leq t \leq a$. If $\lim_{a \rightarrow \infty} L_a = z$, then determine the value of z .

By the arc length formula for a parametrically-defined curve, we have that

$$\begin{aligned} L_a &= \int_0^a \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^a \sqrt{\left(-5e^{-t} \cos(t) - 5e^{-t} \sin(t)\right)^2 + \left(-5e^{-t} \sin(t) + 5e^{-t} \cos(t)\right)^2} dt \\ &= \int_0^a \sqrt{\left(25e^{-2t} \cos^2(t) + 50e^{-2t} \cos(t) \sin(t) + 25e^{-2t} \sin^2(t)\right) + \left(25e^{-2t} \sin^2(t) - 50e^{-2t} \sin(t) \cos(t) + 25e^{-2t} \cos^2(t)\right)} dt \\ &= \int_0^a \sqrt{25e^{-2t} (\cos^2(t) + \sin^2(t)) + 25e^{-2t} (\sin^2(t) + \cos^2(t))} dt = \int_0^a \sqrt{50e^{-2t}} dt = \int_0^a 5e^{-t} \sqrt{2} dt \\ &= -\frac{5\sqrt{2}}{e^a} + 5\sqrt{2} \end{aligned}$$

$$\text{Therefore, } \lim_{a \rightarrow \infty} L_a = \lim_{a \rightarrow \infty} \left(-\frac{5\sqrt{2}}{e^a} + 5\sqrt{2} \right) = 5\sqrt{2}$$

17. Identify each of the following as either an ellipse, hyperbola, or parabola by putting each equation into standard form.

a) $x^2 + y^2 - 10x + 9 = 0$

$$(x^2 - 10x) + y^2 = -9$$

$$(x^2 - 10x + 25) + y^2 = 16$$

$$(x-5)^2 + y^2 = 16$$

This is an ellipse (more technically a circle) with center at $(5, 0)$.

$$\frac{(x-5)^2}{16} + \frac{y^2}{16} = 1$$

$$\text{b) } y^2 + 16x - 4y - 52 = 0$$

$$(y^2 - 4y) + 16x = 52$$

$$(y^2 - 4y + 4) + 16x = 56$$

$$(y - 2)^2 + 16x = 56 \quad \text{This is a parabola that opens to the left.}$$

$$16x = -(y - 2)^2 + 56$$

$$x = -\frac{1}{16}(y - 2)^2 + \frac{7}{2}$$

$$\text{c) } 11x^2 - 25y^2 + 22x + 250y - 889 = 0$$

$$11x^2 - 25y^2 + 22x + 250y - 889 = 0$$

$$(11x^2 + 22x) + (-25y^2 + 250y) = 889$$

$$11(x^2 + 2x) - 25(y^2 - 10y) = 889$$

$$11(x^2 + 2x + 1) - 25(y^2 - 10y + 25) = 889 + 11 - 625 \quad \text{This is a hyperbola that opens left and right.}$$

$$11(x + 1)^2 - 25(y - 5)^2 = 275$$

$$\frac{(x + 1)^2}{25} - \frac{(y - 5)^2}{11} = 1$$